

Name: _____ Tutorial group: _____

Matriculation number:

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18 September 2021 **MH1812 Test 1** 60 minutes

QUESTION 1. (30 marks)

Use mathematical induction to show that

$$5 + 4 + 3 + \cdots + (6 - n) = \frac{1}{2}n(11 - n).$$

whenever n is a positive integer.

[Proof:]

Base case: when $n = 1$, the left-hand side is 5 and the right-hand side is $\frac{1}{2} \cdot 1 \cdot (11 - 1) = 5$. The equality holds.

Inductive step: Suppose that the equality holds for $n = k$, and we shall show that it holds for $n = k + 1$, that is, we shall show that

$$5 + 4 + 3 + \cdots + (6 - k) + (6 - (k + 1)) = \frac{1}{2}(k + 1)(11 - (k + 1)) \tag{1}$$

By induction hypothesis, the LHS of (1) equals

$$\frac{1}{2}k(11 - k) + (5 - k) = \frac{(11k - k^2) + (10 - 2k)}{2} = \frac{-(k^2 - 9k - 10)}{2} = -\frac{(k - 10)(k + 1)}{2},$$

which is exactly the RHS of (1). This completes the proof of the inductive step.

Therefore, by mathematical induction, the original equality holds for all integers $n \geq 1$.

For graders only	Question	1	2(a)	2(b)	3(a)	3(b)	Bonus	Total
	Marks							

QUESTION 2. (30 marks)

(a) (10 points) Find $(18^{12} \cdot 20^{21}) \bmod 60$.

[Solution:] Observe that $18 \cdot 20 \equiv 0 \pmod{60}$, thus

$$18^{12} \cdot 20^{21} \equiv (18 \cdot 20) \cdot (18^{11} \cdot 20^{20}) \equiv 0 \cdot (18^{11} \cdot 20^{20}) \equiv 0 \pmod{60},$$

that is, $(18^{12} \cdot 20^{21}) \bmod 60 = 0$.

(b) Let $\mathbb{R}$ denote the set of reals. For $x, y \in \mathbb{R}$, let $P(x, y)$ denote the predicate “ $x \cdot (y^2 - \frac{1}{1812}) < 0$ ”. What are the truth values of these statements?

(i) (10 points) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, P(x, y)$.

(ii) (10 points) $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, P(x, y)$.

[Solution:]

(i) The statement is false, because $P(0, y)$ is false for all $y \in \mathbb{R}$.

(ii) The statement is false, because $P(x, \frac{1}{\sqrt{1812}})$ is false for all $x \in \mathbb{R}$.

QUESTION 3. (40 marks)

(a) (20 points) Determine whether $p \rightarrow (q \rightarrow (p \wedge q))$ is a tautology.

[Proof:] It is a tautology. One can prove it by manipulating the formula or by a truth table.

$$\begin{aligned} & p \rightarrow (q \rightarrow (p \wedge q)) \\ \equiv & p \rightarrow (\neg q \vee (p \wedge q)) \\ \equiv & p \rightarrow ((\neg q \vee p) \wedge (\neg q \vee q)) \\ \equiv & p \rightarrow ((\neg q \vee p) \wedge T) \\ \equiv & p \rightarrow (\neg q \vee p) \\ \equiv & \neg p \vee (\neg q \vee p) \\ \equiv & T \vee \neg q \\ \equiv & T \end{aligned}$$

p	q	$p \wedge q$	$q \rightarrow (p \wedge q)$	$p \rightarrow (q \rightarrow (p \wedge q))$
T	T	T	T	T
T	F	/	T	T
F	/	/	/	T

(‘/’ denotes an entry that need not be evaluated.)

(b) (20 points) Determine whether the following argument is valid¹.

$$\begin{aligned} & (q \wedge r) \rightarrow \neg p; \\ & q \vee \neg r; \\ & \neg r \rightarrow q; \\ \therefore & \neg p. \end{aligned}$$

[Solution:] The argument is invalid. When $p = T$, $q = T$, $r = F$, all premises are satisfied but the conclusion is not satisfied.

(This is the only counterexample. In a counterexample, p must be true. The second premise is $r \rightarrow q$, combining with the third premise $\neg r \rightarrow q$, yields that q must be true. The first premise then becomes $r \rightarrow F$, which is true only when r is false.)

BONUS QUESTION. (10 marks)

[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]

Prove that $(5^{69} - 1)/4$ is odd and composite. (Recall that a positive integer which can be factored into smaller positive integers, neither of which is one, is a composite.)

[Proof:] In general we show that (i) $(5^n - 1)/4$ is odd if $n > 1$ is odd and (ii) $(5^n - 1)/4$ is composite if $n = 3m$ for some odd integer m .

For part (i), note that $5^2 \equiv 1 \pmod{8}$ and so $5^n \equiv (5^2)^{(n-1)/2} \cdot 5 \equiv 5 \pmod{8}$, hence $5^n = 8k + 5$ for some integer k and thus $(5^n - 1)/4 = 2k + 1$ is odd.

For part (ii), note that $5^n - 1 = (5^m)^3 - 1 = (5^m - 1)(5^{2m} + 5^m + 1)$. Applying part (i) to $(5^n - 1)/4$, we know that it is an integer, thus $(5^n - 1)/4$ is an integer multiple of $(5^m - 1)/4$.

(In fact, using the identity that $x^k - y^k = (x - y)(x^{k-1} + x^{k-2}y + \cdots + y^k)$, one can show that $(5^m - 1)/4$ is a divisor of $(5^n - 1)/4$ if m is an odd divisor of n .)

¹The inference rules you may use are: modus ponens, modus tollens, conjunctive simplification, conjunctive addition, disjunctive addition, disjunctive syllogism, hypothetical syllogism, disjunction elimination and rule of contradiction.